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BUSINESS ANALYTICS

Case Study Report

Analysis of Student Willingness-to-Pay for Premium Ed-Tech Tools: A Predictive Modelling Approach

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Abstract

Ed-Tech platforms increasingly use freemium business models in which basic functionality is provided at no cost while advanced features are offered through premium subscriptions. Although these models attract large numbers of users, identifying users who demonstrate meaningful interest in premium features remains a business analytics challenge. This study investigates review-based signals associated with premium engagement in Ed-Tech applications using publicly available Google Play reviews.

A dataset of 10,800 publicly available reviews from Duolingo, Photomath, and Brainly was collected through web scraping. Each application contributed 3,600 reviews. The dataset contains review ratings, review text, application information, helpfulness indicators, review dates, application versions, and other review metadata. Duplicate records were checked and review-level textual indicators were engineered to identify references to premium subscriptions, purchasing, pricing, free access, cancellation, and feature limitations.

A review-derived premium engagement score was created as a proxy measure rather than as a direct observation of willingness-to-pay or actual subscription conversion. Linear Regression, Decision Tree Regression, and Random Forest Regression were developed using review rating, review length, word count, helpfulness, and application as predictors. Linear Regression obtained an MAE of 0.1258, RMSE of 0.2886, and R^2 of 0.1308. Decision Tree Regression obtained an MAE of 0.1124, RMSE of 0.2923, and R^2 of 0.1087, while Random Forest obtained an MAE of 0.1146, RMSE of 0.2947, and R^2 of 0.0936.

The findings indicate that public review metadata can provide useful exploratory insights into premium-related user discussions, but the available variables explain only a limited portion of variation in the proxy engagement score. The study therefore highlights the value of combining review analytics with transaction, demographic, subscription, and behavioural data for more direct future conversion modelling.

1 Problem Statement and Objectives

1.1 Problem Statement

Ed-tech platforms attract millions of users through freemium models, but only a small percentage convert to premium subscriptions, resulting in inefficient marketing efforts and lost revenue opportunities. Businesses need a data-driven approach to identify users with genuine purchase intent rather than relying solely on basic engagement metrics.

This study uses publicly available app-review text and platform-level review metadata to identify observable signals associated with premium engagement while recognising that verified behavioural, attitudinal, demographic, and transaction-level information is unavailable.

Since the collected dataset does not contain verified subscription transactions, demographic information, or direct willingness-to-pay responses, the analysis focuses on review-derived premium engagement signals as a proxy.

1.2 Objectives

The main objectives of the case study are:

1. To collect and analyse a large publicly available dataset of Ed-Tech application reviews.
2. To identify review-based signals associated with premium features, pricing, subscriptions, free access, cancellation, and feature limitations.
3. To develop and evaluate predictive models for the review-derived premium engagement score.
4. To derive practical business insights and recommendations for premium-feature and subscription strategy.

2 Data Collection and Dataset Description

2.1 Data Source

The dataset consists of publicly available Google Play reviews from three Ed-Tech applications:

- Duolingo
- Photomath
- Brainly

The reviews were collected through web scraping of publicly accessible Google Play review information. A Google Play review scraping workflow was used to collect structured review records without relying on a pre-existing Kaggle, UCI, or GitHub dataset.

The collected records contain fields such as review identifier, rating, review text, review date, helpfulness count, application version, application identifier, application title, language, and country.

2.2 Data Collection Procedure

The data collection procedure consisted of the following steps:

1. Public Google Play review pages were selected for Duolingo, Photomath, and Brainly.
2. A web-scraping workflow was configured to collect publicly available review records.
3. Reviews were collected with English language and United States review settings.
4. A target of 3,600 reviews was collected for each application.
5. The three application datasets were combined into a single dataset.
6. Review identifiers were checked for duplicate records.
7. The final dataset contained 10,800 review records.

2.3 Dataset Distribution

The final dataset contains an equal number of reviews for each application.

Table 1: Distribution of Reviews by Application

Application	Reviews	Percentage
Duolingo	3,600	33.33%
Photomath	3,600	33.33%
Brainly	3,600	33.33%
Total	10,800	100%

2.4 Dataset Attributes

The main attributes used in the analysis are:

Table 2: Important Dataset Attributes

Attribute	Description
reviewId	Unique identifier associated with the review.
score	User rating assigned to the application, from 1 to 5.
text	Textual content of the user review.
thumbsUp	Helpfulness count associated with the review.
reviewedAt	Date and time associated with the review.
appVersion	Application version when available.
appId	Application identifier.
appTitle	Application name.
language	Language associated with the collected review.
country	Country setting used during collection.

2.5 Dataset Limitations

The dataset represents public app-review activity and does not provide verified student identities, demographic characteristics, subscription transactions, actual payment information, or confirmed willingness-to-pay values.

Therefore, the analysis does not claim that every reviewer is a student or that a premium-related review necessarily resulted in a purchase. Review language is instead used to construct an observable proxy for premium engagement.

3 Data Preparation and Exploratory Analysis

3.1 Data Cleaning

The following preprocessing steps were performed:

- Duplicate review records were checked using review identifiers.
- Review text was converted to a consistent textual format.
- Missing textual values were handled during feature extraction.
- Review character length and word count were calculated.
- Numerical variables were inspected before modelling.
- Premium-related keyword indicators were generated.

The final analytical dataset contained 10,800 review records.

3.2 Rating Distribution

The distribution of review ratings was:

Table 3: Review Rating Distribution

Rating	Count	Percentage
1	1,540	14.26%
2	285	2.64%
3	478	4.43%
4	1,194	11.06%
5	7,303	67.62%
Total	10,800	100%

The average rating across the dataset was approximately 4.15.

3.3 Review Length Analysis

Review text length was calculated using the number of characters and words.

The mean review length was approximately 55.23 characters, while the median was 27 characters. The distribution indicates that many reviews are short, although a smaller number of reviews contain substantially more detailed feedback.

3.4 Premium-Related Feature Engineering

A direct purchase-intent label was not used because explicit purchase-related language is relatively sparse in public reviews. Instead, a broader review-derived premium engagement score was constructed.

Six categories of premium-related signals were extracted:

- Premium and subscription references
- Purchase and payment references
- Price and affordability references
- Free access and advertisement references
- Cancellation and refund references
- Feature limitation and locked-feature references

Binary indicators were created using keyword-based text matching.

Table 4: Premium-Related Text Signals

Feature	Reviews Containing Signal
has premium	115
has purchase	210
has price	42
has free	300
has cancel	24
has limitation	33

The premium engagement score was calculated as:

$$\text{Premium Engagement Score} = \text{Premium} + \text{Purchase} + \text{Price} + \text{Free} + \text{Cancel} + \text{Limitation}$$

The resulting score was divided into three engagement levels:

- Low: score = 0
- Medium: score = 1
- High: score ≥ 2

3.5 Premium Engagement Distribution

The final engagement distribution was:

Table 5: Premium Engagement Distribution

Engagement Level	Count	Percentage
Low	10,236	94.78%
Medium	438	4.06%
High	126	1.17%
Total	10,800	100%

The mean review-derived premium engagement score was approximately 0.067.

3.6 Application-Level Analysis

The application-level analysis showed different levels of review-derived premium engagement.

Table 6: Application-Level Premium Engagement

Application	Reviews	Avg. Score	High
Brainly	3,600	0.0275	11
Duolingo	3,600	0.0769	49
Photomath	3,600	0.0967	66

The corresponding high-engagement proportions were approximately:

- Brainly: 0.31%
- Duolingo: 1.36%
- Photomath: 1.83%

These values represent review-derived engagement and should not be interpreted as actual premium conversion rates.

4 Analytics Method and Implementation

4.1 Analytical Objective

The predictive modelling component estimates the review-derived premium engagement score using observable review metadata and application information.

The target variable is:

$$\text{Target} = \text{Premium Engagement Score}$$

The target is a proxy constructed from review language and should not be interpreted as an observed subscription transaction or direct willingness-to-pay measure.

4.2 Predictor Variables

The following predictor variables were used:

- Review rating
- Review length
- Word count
- Thumbs-up count
- Application

The keyword indicators used to construct the target were excluded from the predictive feature set to avoid direct target leakage.

4.3 Train-Test Split

The dataset was divided into training and testing subsets using an 80:20 split.

Table 7: Train-Test Split

Dataset	Records
Training set	8,640
Testing set	2,160
Total	10,800

A random state of 42 was used to make the split reproducible.

4.4 Preprocessing

Numerical features were processed using median imputation.

The application variable was processed using:

- Most-frequent-value imputation
- One-hot encoding

The preprocessing steps were implemented using a Scikit-learn pipeline.

4.5 Predictive Models

Three regression models were evaluated:

1. Linear Regression
2. Decision Tree Regression
3. Random Forest Regression

4.6 Evaluation Metrics

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and coefficient of determination (R^2).

4.6.1 Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE measures the average absolute difference between actual and predicted values.

4.6.2 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE gives greater weight to larger prediction errors.

4.6.3 Coefficient of Determination

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

R^2 indicates the proportion of variation in the target explained by the model.

5 Comparison with Recent Published Studies

Recent research demonstrates different approaches for understanding willingness-to-pay, subscription-payment intention, and student perspectives toward digital education.

Table 8: Comparison with Recent Published Studies

Study / Year	Dataset	Method	Metric	Key Result	Comparison
Tyrväinen & Karjaluoto (2024)	55 studies and 15 qualitative interviews on freemium services	Mixed-method and meta-analysis	Effect sizes and qualitative analysis	Functional, hedonic, social and price value were identified as predictors of willingness-to-pay. Trust also mediated relevant relationships.	The present study also examines freemium-related behaviour, but uses large-scale public app reviews and a review-derived proxy.
Veguru, Naren & Singam (2024)	Questionnaire responses from college students regarding online education	Machine-learning classification	Classification accuracy	Random Forest achieved a reported accuracy of 93% among the tested classification approaches.	Uses student-focused educational data and ML classification, whereas this study uses public app reviews and regression modelling.
Liu, Park & Wang (2025)	274 ChatGPT user responses	Regression and moderated mediation	Regression and mediation analysis	Perceived usefulness positively influenced satisfaction and subscription-payment intention, with satisfaction partially mediating the relationship.	Examines subscription-payment intention using survey data, whereas this study uses 10,800 public reviews and a text-derived proxy.

5.1 Comparison with the Present Study

The present study differs from the compared studies mainly in its data source, target definition, and analytical approach.

Instead of questionnaire responses or verified subscription transactions, this study uses 10,800 publicly available Google Play reviews from Duolingo, Photomath, and Brainly.

The study constructs a review-derived premium engagement score from observable references to premium subscriptions, payment, pricing, free access, cancellation, and feature limitations.

This score provides a scalable mechanism for analysing public user feedback, but it is not equivalent to observed subscription purchase behaviour or directly measured willingness-to-pay.

Unlike studies using survey or learner-level behavioural information, the present models use basic review metadata and text-derived characteristics while excluding the keyword indicators used to construct the target.

The relatively limited R^2 values indicate that the available predictors explain only a small portion of variation in the proxy target. This highlights the potential value of richer behavioural, demographic, survey, and transaction-level information for direct prediction of subscription payment or willingness-to-pay.

6 Results, Business Insights and Recommendations

6.1 Model Results

The predictive modelling results are shown below.

Table 9: Predictive Model Performance

Model	MAE	RMSE	R^2
Linear Regression	0.1258	0.2886	0.1308
Decision Tree	0.1124	0.2923	0.1087
Random Forest	0.1146	0.2947	0.0936

The models show relatively limited explanatory power for the review-derived premium engagement score. The differences in performance should therefore be interpreted in the context of the proxy nature of the target.

6.2 Model Error Comparison

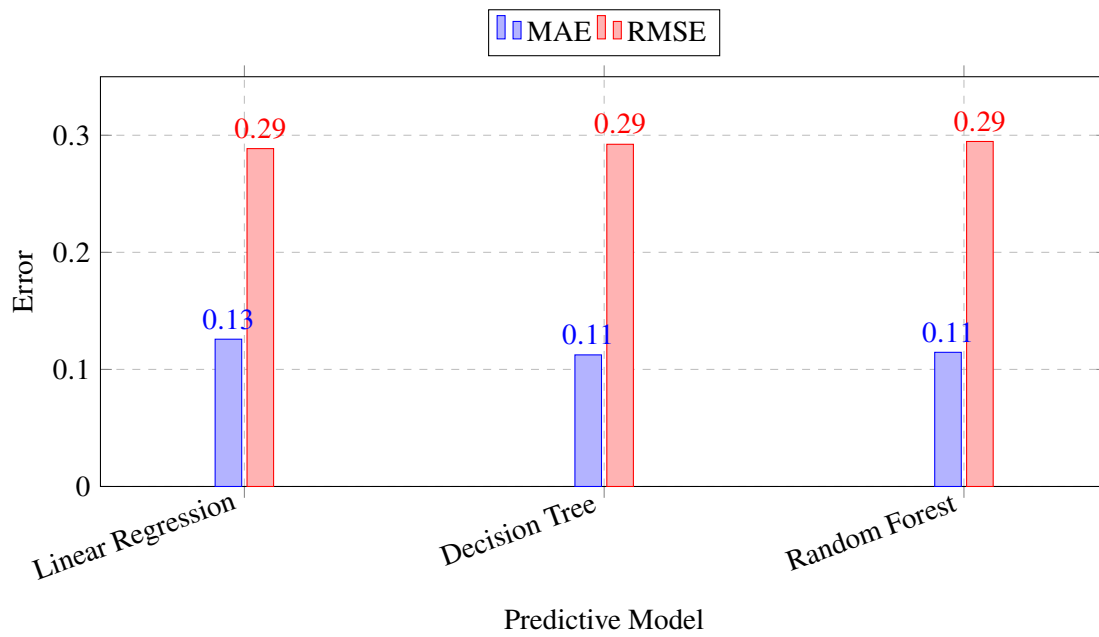


Figure 1: Comparison of MAE and RMSE across predictive models

6.3 R^2 Comparison

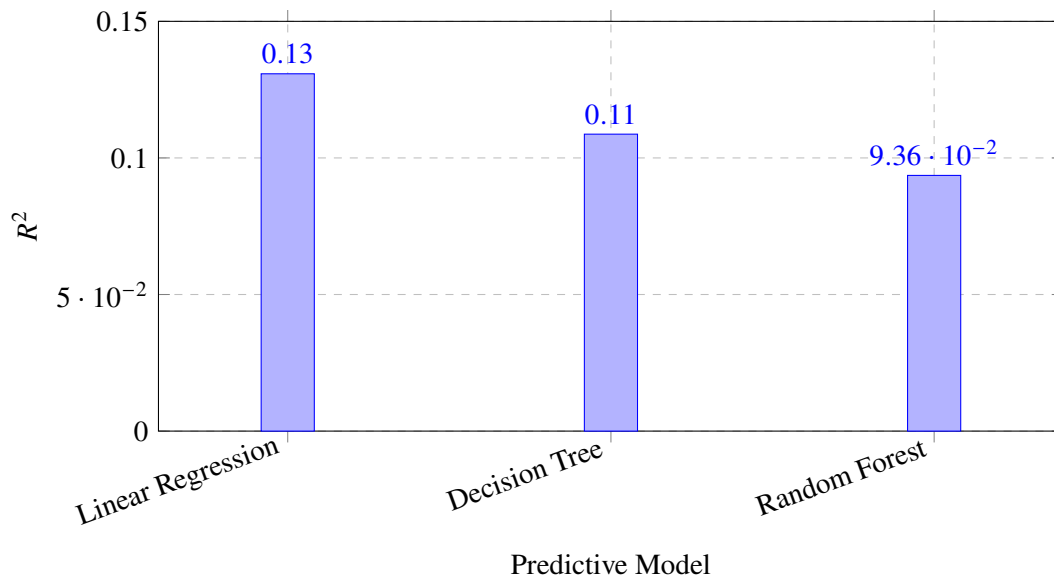


Figure 2: R^2 comparison across predictive models

6.4 Model Interpretation

The three regression models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 .

Linear Regression produced an MAE of 0.1258, RMSE of 0.2886, and R^2 of 0.1308. Decision Tree produced an MAE of 0.1124, RMSE of 0.2923, and R^2 of 0.1087, while Random Forest produced an MAE of 0.1146, RMSE of 0.2947, and R^2 of 0.0936.

The results indicate that the available review metadata and basic text-length characteristics explain only a limited portion of the variation in the review-derived premium engagement score.

The target itself is a proxy constructed from review language rather than observed subscription transactions. Therefore, the models should be interpreted as exploratory predictive models rather than direct predictors of actual willingness-to-pay or subscription conversion.

6.5 Business Insights

The exploratory analysis provides several observations relevant to Ed-Tech subscription and premium-feature strategy.

- Premium-related engagement is concentrated in a small proportion of reviews. The dataset contains 10,800 reviews, with 126 classified as High and 438 classified as Medium.
- The three applications show different levels of review-derived premium engagement. Brainly has an average score of 0.0275, Duolingo 0.0769, and Photomath 0.0967.
- Photomath has the highest average review-derived premium engagement score in the collected dataset, while Brainly has the lowest.
- Premium-related review language includes references to subscriptions, payment, pricing, free access, advertisements, cancellation, and feature limitations.
- The predictive models produced relatively limited R^2 values, indicating that the basic review-level variables do not explain most variation in the proxy target.

From a business analytics perspective, the findings suggest that public review data can be used as an exploratory feedback-monitoring mechanism. However, transaction-level subscription data and verified user characteristics would be required for reliable individual-level conversion prediction.

6.6 Business Recommendations

Based on the exploratory analysis and predictive modelling results, the following recommendations can be considered by Ed-Tech platforms:

1. Monitor premium-related user feedback

Platforms can systematically analyse public reviews for references to subscriptions, pricing, advertisements, feature restrictions, and premium access. This can help identify recurring monetization-related concerns.

2. Investigate application-specific premium feedback

The three applications showed different levels of review-derived premium engagement. Platforms should therefore analyse application-specific feedback rather than assuming identical premium-related behaviour across products.

3. Examine pricing and feature-access discussions

References to price, free access, advertisements, cancellation, and feature limitations can be monitored as recurring themes in user feedback. These signals can support product and monetization discussions.

4. Combine review data with behavioural data

The relatively limited predictive performance indicates that review data alone is insufficient for direct conversion prediction. Combining review signals with usage frequency, subscription history, feature usage, demographics, and transaction information could provide a stronger analytical basis.

5. Use predictive models as decision-support tools

The models developed in this study should be treated as exploratory analytical tools rather than direct predictors of actual willingness-to-pay or subscription conversion. Business decisions should incorporate additional verified customer and transaction-level information.

7 Conclusion and References

7.1 Conclusion

This study analysed 10,800 publicly available Google Play reviews from Duolingo, Photomath, and Brainly to investigate user discussions related to premium features, pricing, and subscription experiences.

The exploratory analysis identified premium-related language involving subscriptions, payment, pricing, free access, cancellation, and feature limitations. The three applications showed different levels of review-derived premium engagement, with average scores of 0.0275 for Brainly, 0.0769 for Duolingo, and 0.0967 for Photomath.

Three regression models were developed using review rating, review length, word count, helpfulness, and application as predictors of the review-derived premium engagement score. Linear Regression achieved an R^2 of 0.1308, Decision Tree achieved an R^2 of 0.1087, and Random Forest achieved an R^2 of 0.0936.

The results indicate that basic review metadata and text characteristics explain only a limited portion of variation in the proxy target.

The study demonstrates that publicly available review data can provide useful exploratory insights for Ed-Tech subscription and premium-feature analysis. However, the premium engagement score is a proxy and should not be interpreted as actual willingness-to-pay or confirmed subscription conversion.

Future work could incorporate verified transaction data, subscription history, usage frequency, user demographics, feature usage, and other behavioural variables to develop more direct conversion prediction models.

Overall, the analysis provides a data-driven framework for monitoring premium-related user feedback and identifying themes that can support future pricing, feature-access, and customer-engagement decisions.

7.2 References

1. Tyrväinen, O., & Karjaluo, H. (2024).
Willingness to pay for freemium services: Addressing the differences between monetization strategies.
International Journal of Information Management, 77, 102787.
DOI: 10.1016/j.ijinfomgt.2024.102787.
2. Veguru, H. S. K., Naren, J., & Singam, Y. (2024).
Student's Interest and Opinion Towards Online Education.
Procedia Computer Science, 233, 590–596.
DOI: 10.1016/j.procs.2024.03.248.
3. Liu, Y., Park, Y., & Wang, H. (2025).
The mediating effect of user satisfaction and the moderated mediating effect of AI anxiety on the relationship between perceived usefulness and subscription payment intention.
Journal of Retailing and Consumer Services, 84, 104176.
DOI: 10.1016/j.jretconser.2024.104176.
4. Google Play. (2026).
Public user reviews for Duolingo, Photomath and Brainly collected through web scraping for this study.
5. Apify. (2026).
Google Play Store Reviews Scraper.
Public web-scraping workflow used for collecting the review dataset.

A Feature Summary

The main engineered features used in the analysis are:

Table 10: Feature Engineering Summary

Feature	Description
review length	Number of characters in the review text.
word count	Number of words in the review text.
has premium	Indicates premium/subscription-related terms.
has purchase	Indicates purchase/payment-related terms.
has price	Indicates pricing or affordability terms.
has free	Indicates free-access, advertisement, or freemium terms.
has cancel	Indicates cancellation, refund, or unsubscribe terms.
has limitation	Indicates limitation, restriction, or locked-feature terms.
premium engagement score	Sum of the six premium-related binary indicators.
premium engagement level	Low, Medium, or High based on the engagement score.

B Reproducibility Structure

The project is organised as follows:

```

EdTech_Case_Study/
|
|-- README.md
|
|-- data/
|   |-- raw/
|   |   '-- google_play_reviews_raw.csv
|   |
|   '-- processed/
|       '-- edtech_reviews_analysis.csv
|
|-- notebooks/
|   '-- analysis.ipynb
|
'-- report/
    '-- CB.SC.U4CSE23039_BA_CS.pdf

```